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Sprint-Nav DP - A New DP Reference Sensor for Shallow Water Operations

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Abstract

SPRINT-Nav DP is Sonardyne's latest position reference system (PRS), purpose-built to enhance dynamic positioning (DP) operations in shallow water environments up to 230 metres water depth such as the continental shelf, inshore waters and rivers and estuaries. Designed to meet the stringent redundancy requirements of DP2 and DP3-class vessels, SPRINT-Nav DP provides a reliable alternative or complement to Global Navigation Satellite Systems (GNSS), offering distinct operational advantages over traditional reference sensors.

This paper explores the challenges faced by DP vessels operating in shallow offshore environments, where sensor masking, infrastructure clutter, and GNSS vulnerabilities can compromise DP capability. It presents the underlying technology of SPRINT-Nav DP, including its integrated inertial and Doppler velocity capabilities, and outlines how these features contribute to system robustness.

A case study is included, showcasing SPRINT-Nav DP deployed on a DP2 heavy-lift vessel operating at its maximum rated altitude, demonstrating the system's performance and reliability in real-world conditions. Additionally, the paper highlights value-added features—such as real-time water current profiling and spoofing detection capabilities—not traditionally offered by DP reference sensors, and discusses how these can be leveraged to improve situational awareness and operational efficiency in offshore projects.

Introduction

The rapid expansion of offshore renewable energy projects has led to a significant increase in the use of DP2 and DP3-class vessels operating in shallow water environments. Over the past decade, vessel designs have evolved to meet the specific demands of these conditions. However, shallow water operations pose a unique set of challenges compared to deep water activities, including increased infrastructure, reduced effectiveness of DP sensors, and external influences.

Despite these operational challenges, the technologies commonly used to provide robust and reliable position references for dynamic positioning (DP) systems have remained largely unchanged. Shallow water operations tend to be shorter in duration, and more dynamic than deep water, making deployment of instruments on the seabed impractical.

This paper examines the limitations of conventional Position Reference Systems (PRSs) in shallow water and introduces Sonardyne's SPRINT-Nav DP—a hybrid acoustic-inertial navigation system specifically designed to address these challenges.

SPRINT-Nav DP enables position resilience without the need for deployed subsea or surface targets, thereby reducing system complexity and risk. Additionally, it incorporates real-time water speed profiling within a single integrated unit, offering enhanced situational awareness and operational capability for shallow water DP operations.

Position Reference Sensor Overview

Hydroacoustic Position Reference (HPR)

Hydroacoustic Position Reference Systems (HPRs), such as USBL (Ultra-Short Baseline) and LBL (Long Baseline), are widely used in dynamic positioning (DP), particularly for subsea operations. However, in shallow water environments, HPR systems face significant operational challenges that can affect both their reliability and accuracy.

These systems typically require an acoustic transceiver to be statically deployed from the vessel, along with at least one acoustic beacon placed on the seabed to act as a fixed reference point for DP operations. The deployment and recovery of the beacon(s) can be timeconsuming and may require ROV intervention.

In some operations—such as pipeline surveys—a mobile position reference is required. In this mode, a mobile beacon is attached to a subsea asset (ROV), allowing the DP desk to track the asset's position in real-time. This enables the HPR system to provide a position reference during these operations.

The HPR system functions by transmitting an acoustic interrogation pulse from the vessel's transceiver to the subsea beacon. The beacon responds with a return signal, which is received at the transceiver. The system measures the azimuth and elevation of the incoming signal, from which a relative or absolute position can be calculated.

As the acoustic signals propagate through the water column, they are vulnerable to localised interference. Vessel-generated and surface noise can mask the signal, especially in shallow water where thruster output must be increased to counteract fast-moving surface currents and wave action. Acoustic signals are also subject to multipath effects, caused by reflections off hard surfaces—an issue particularly relevant when operating within wind farms. In addition, stratification (layering of the water column due to varying temperature or salinity) can impede acoustic signal propagation, either preventing it from reaching the seabed or returning to the surface. While modern HPR systems can compensate for refraction caused by this layering, at high elevation angles the signal may be reflected entirely off one of these layers, preventing successful reception at the surface. Layering in shallow water can also change quickly and become chaotic due to the vessel's own thruster activity.

HPR systems also require a clear line of sight between the vessel transceiver and the subsea beacon, which can restrict vessel manoeuvrability. Despite HPR systems providing a extremely robust solution for deep water operations, one of the main constraints of using HPR systems in shallow water is the limited operational footprint they provide. Due to the factors outlined above, HPR systems typically deliver a robust position reference—within 3 metres of positional error—up to a few hundred metres from the beacon in shallow water. While techniques such as depth aiding can be employed to expand this operational footprint, the effectiveness of these techniques is limited.

Laser

Laser-based PRS's—including laser scanners and laser reflectors—are frequently employed in DP applications for offshore vessels, particularly when operating in close proximity to fixed structures such as oil platforms or wind turbines. Due to the wide adoption of these systems as PRS's, many fixed installations have pre-installed reflectors, offering a quick, reliable PRS without the need to deploy additional hardware.

While these systems offer high-precision measurements within relatively short ranges, they are subject to a number of operational and technical limitations.

The typical operational range for a laser-based PRS is between 10 and 1000 metres; however, this range can be significantly diminished under adverse environmental conditions. Atmospheric factors such as fog, mist, rain, snow, sea spray, dust, and smoke can severely attenuate the laser signal, leading to reduced system performance or complete signal loss. Additionally, continuous line-of-sight between the laser unit and the target—whether a reflective structure or a dedicated reflector—is essential, thereby imposing constraints on vessel manoeuvrability in the vicinity of structures.

In cases where dedicated laser reflectors are installed, these must be regularly inspected and maintained to ensure operational reliability. The performance and accuracy of laser-based PRS's are therefore heavily dependent on both environmental stability and the condition of the reflective infrastructure.

Radio

Radio-based PRS's provide an extended operational range—typically exceeding 1,000 metres—when compared to laser-based alternatives, and are generally less affected by environmental conditions such as fog, rain, and sea spray. However, despite these advantages, radio-based systems also present limitations that may impact their suitability for certain DP operations.

A key operational constraint is that radio-based systems require the installation of transponders or active targets on the reference structure, such as a platform or Floating Production Storage and Offloading (FPSO) unit. This requirement makes them less suitable for short-term or temporary operations, where rapid deployment and minimal infrastructure are preferred.

Although more resilient than laser-based systems in the face of environmental interference, radio-based PRSs remain susceptible to signal degradation during severe weather conditions, particularly during electrical storms. Additionally, these systems operate within a defined frequency spectrum, typically in the 3-27 GHz range, which can become congested in offshore environments—especially in proximity to rigs and vessels equipped with extensive communication systems.

As such, while radio-based PRSs offer clear benefits in terms of range and robustness, careful planning and frequency management are required to ensure reliable performance in complex offshore scenarios.

Global Navigation Satellite Systems (GNSS)

Global Navigation Satellite Systems (GNSS) are widely adopted as the primary Position Reference System (PRS) on most vessels, particularly those operating with Dynamic Positioning (DP) capabilities. The technology is well-established in the maritime industry, with Differentially Corrected GNSS (DGNSS) systems routinely achieving positional accuracies in the range of 0.5 to 1 metre.

The use of multiple satellite constellations—such as GPS, GLONASS, Galileo, and BeiDou—provides both global coverage and redundancy, enhancing operational reliability in regions where certain constellations may be more favourable than others. However, effective GNSS performance requires an unobstructed line of sight between the antennae and the sky. In offshore environments, this can be compromised by physical obstructions including platform superstructures, cranes, adjacent vessels, fjords, and harbour walls.

In addition to physical masking, GNSS signals are vulnerable to environmental interference, particularly atmospheric disturbances. This is especially problematic at high latitudes and during periods of increased solar activity, where signal degradation due to scintillation is more pronounced. When correction data is lost—due to signal interruption or obstruction—the positional accuracy of DGNSS can degrade significantly, often falling to 5-10 metres, compared to the typical 0.5-1 metre performance required for DP.

A growing concern is the intentional disruption of GNSS signals. Signal jamming—which blocks all incoming satellite data—and spoofing, where manipulated signals are used to mislead the receiver into reporting a false position, are now recognised as credible threats in both commercial and defence sectors.

Such interference not only impacts vessel positioning but also affects onboard systems dependent on GNSS-derived timing, potentially compromising broader navigational and operational functions.

Taut wire

Taut wire systems are a traditional form of PRS used in DP operations. These systems operate by deploying a wire connected to a clump- weight from the vessel to the seabed. By continuously monitoring the angle and length of the wire, the system calculates the vessel's position relative to the fixed location of the clump weight on the seabed.

Taut wire systems typically have a maximum wire length of 400 metres, which directly limits the vessel's operational footprint. For example, in 100 metres of water depth, the maximum horizontal range from the clump weight is approximately 387 metres. This constraint can restrict vessel manoeuvrability, particularly in operations requiring greater lateral flexibility.

A critical operational consideration is that the wire must remain free from snagging. Any entanglement can disrupt system functionality, degrade positioning accuracy, and potentially cause damage to subsea infrastructure. There have been documented incidents where the clump weight was unintentionally dragged along the seabed, rendering the system ineffective and increasing the risk of damage to sensitive subsea assets. This is of particular concern in regions with dense subsea infrastructure, such as the Middle East.

In shallow water, taut wire systems are further challenged by wave-induced vessel motion, especially vertical motion due to heave. These dynamic forces can lead to rapid fluctuations in wire angle and tension, adversely affecting measurement accuracy. In cases where vessel heave exceeds the capacity of the system's heave compensation mechanism, the wire may become slack and hang vertically in the water column. This condition significantly degrades the system's ability to provide a reliable position solution and may lead to false position data or signal dropouts.

While taut wire systems have historically played a key role in DP operations, their limitations in shallow and structure-dense environments, combined with the mechanical risks associated with seabed contact, taut wire systems are being used less for DP operations.

SPRINT-Nav DP - An alternative to existing technologies

For over 25 years, Sonardyne has been a leading provider of Ultra-Short Baseline (USBL) tracking, positioning, and telemetry systems, along with Class-certified deployment solutions, for use as a PRS in DP operations. Building on this foundation, Sonardyne introduced its first tightly integrated acoustic and inertial navigation systems to the DP market in 2011, further enhancing the reliability and accuracy of subsea and surface navigation.

Drawing on this extensive heritage, SPRINT-Nav DP has been developed to deliver accurate, precise, and robust navigation and guidance data for both subsea and surface vessels. It is the result of decades of technological advancement and operational experience in challenging offshore environments.

Inertial Navigation Systems (INS), while highly valuable for subsea navigation, are inherently susceptible to positional drift over time when external corrections are unavailable. This drift arises due to the accumulation of small sensor errors, such as gyro bias and accelerometer noise. To mitigate this, DVL-INS (Doppler velocity log - inertial navigation systems) systems rely on external aiding inputs—primarily bottom-lock velocity measurements from the DVL. Vehicle dynamics (such as turns, loops, or speed changes) are also beneficial to aid the INS, although not essential for operation. The aiding data is used to constrain the INS error growth and correct the navigation solution.

In marine operations, particularly in DP scenarios, vessel movement is often restricted. Vessels are typically required to hold station over a fixed point, yaw slowly to weather, or transit short distances, which limits the amount of kinematic excitation available to aid the INS. These low-dynamic conditions can challenge traditional INS performance due to fewer opportunities for heading changes, which are critical for error correction.

Combining DVL with INS helps to prevent this with the DVL providing velocity information to the INS, reducing the drift rate. The use of Sonardyne's Syrinx DVL provides ultra-tight coupling of the DVL and INS data allowing for increased robustness and error management making the system ideal for DP operations. Compared to stand alone units, The INS utilises each of the DVL's four beams independently, requiring only two of the four DVL transducers to return good signals to continue to operate. This adds a layer of resilience not always available with stand-alone units.

As with all products in the SPRINT-Nav family, the sensor fusion architecture of SPRINT-Nav DP intelligently leverages data from all internal sensors to ensure seamless integration and continuous high-accuracy positioning. This results in unprecedented levels of performance compared to standalone instruments, particularly in environments where conventional PRS's face limitations due to signal interference, masking, or operational constraints.

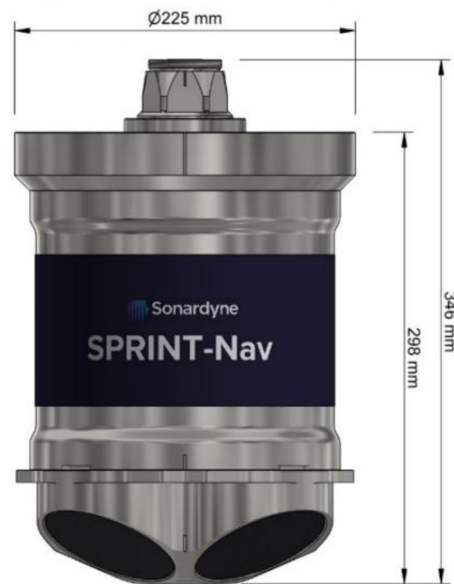


Figure 1—Sonardyne's SPRINT-Nav DP

SPRINT-Nav DP provides latitude and longitude updates via a NMEA-0183 GLL telegram compatible with DP systems. It is also possible to output additional telegrams to other systems on the vessel, for example a pseudo GGA message useful for survey operations, or HDT for systems requiring heading data. All SPRINT-Nav DPs come equipped with highly accurate gyroscopes and accelerometers, which are not affected by magnetism and provide a true north seeking gyrocompass. An internal battery provides ride through during periods of power outages for up to 5 minutes. This eliminates the need for a settling period after power outages. SPRINT-Nav DP offers an easy-to-use Web UI, which provides an intuitive dashboard viewer as well as configuration and detailed status pages for integration and operation.

SPRINT-Nav DP uses the seabed as its reference, without the need to deploy a transponder, or a clump weight. SPRINT-Nav DP emits an acoustic pulse via the DVL component, which is reflected off the seabed and received by the unit on the vessel. SPRINT-Nav DP is therefore ideal as a DP reference when there are no other assets available to act as references, a vessel is unable to deploy anything on the seabed, and when the vessel needs an operational footprint greater than can be offered by HPR and taut wire. An initial vessel position is needed to localise the system, after which point the GNSS can be disabled from use, and the system operates independently from any other position sources. In some instances, the GNSS signal can be generated from non-GNSS technology, for example using an HPR system to provide an initial vessel location. This enables SPRINT-Nav DP to be used in many areas of the world where GNSS is susceptible to interference.

Due to the drift rate of the DVL-INS solution, there will come a point when the position solution has diverged away from the other PRS's. At this point, the system can be reset by enabling the GNSS input feed. The amount of accepted divergence is a user configurable field, allowing DP operators to adjust the acceptable limits of the systems performance to suit the vessels operation.

Installation of SPRINT-Nav DP must ensure line-of-sight to the seabed. As the system is less susceptible to vessel borne noise than some other acoustic position reference sensors, the deployment can be less deep compared to USBL systems. A suitable installation could be compared to a vessel's doppler speed log sensor. The system is optimised for vessel installations using a rugged titanium housing and single cable connection.

GNSS Spoofing

GNSS spoofing is an escalating global concern that poses a serious threat to the safety, reliability, and continuity of marine operations. This form of interference involves the deliberate transmission of false satellite signals to deceive receivers, causing them to report inaccurate position, velocity, or time data. While previously considered a niche or theoretical risk, spoofing is now being routinely observed in several global hot-spots, with increasing frequency and sophistication.

One notable example is the Southern Baltic Sea, a region with a high concentration of offshore renewable energy developments. Here, GNSS spoofing events are occurring daily, disrupting critical marine operations including vessel navigation, survey, and construction activities¹. These disruptions not only delay project timelines but also pose significant safety risks for personnel and assets operating in the affected areas.

In this context, Sonardyne's SPRINT-Nav DP system offers a valuable and effective tool for detecting and mitigating the impact of GNSS spoofing. The system architecture allows the GNSS signal to be maintained as a live input, even when it is not being used in the active navigation solution. This enables real-time visual monitoring of both the SPRINT-Nav DP position and the GNSS-derived position, making it possible for operators to identify deviations and assess data integrity during operations. By providing this independent means of position validation, SPRINT-Nav DP enhances situational awareness, supports decision-making, and helps operators maintain control even in environments where GNSS reliability is compromised.

This capability is particularly beneficial in high-risk zones or where DP operations are class-critical, offering an additional layer of resilience, redundancy, and operational assurance against emerging threats in the global maritime domain.

Water Current Profiling

In addition to SPRINT-Nav DP providing a high-accuracy independent position, it can simultaneously also provide water current profiling information. The system operates as both a DVL to aid the INS for position calculations, and also as an Acoustic Doppler Current Profiler (ADCP) which measures suspended particle speeds and hence the water current speed. The ADCP function requires stand-alone software (Origin Viewer as shown in [Figure 2](#)) included with SPRINT-Nav DP for real-time visualisation of the water column speeds. The data can also be output as a PD0 message for logging on a 3rd party system.

The ability to measure position and water current means that SPRINT-Nav DP can provide motion and speed compensated water current measurements. This feature of the system can be used to determine when safe to deploy equipment and when a vessel is safe to operate. It can also be used to verify local environmental models for water current, e.g. DP water current model.

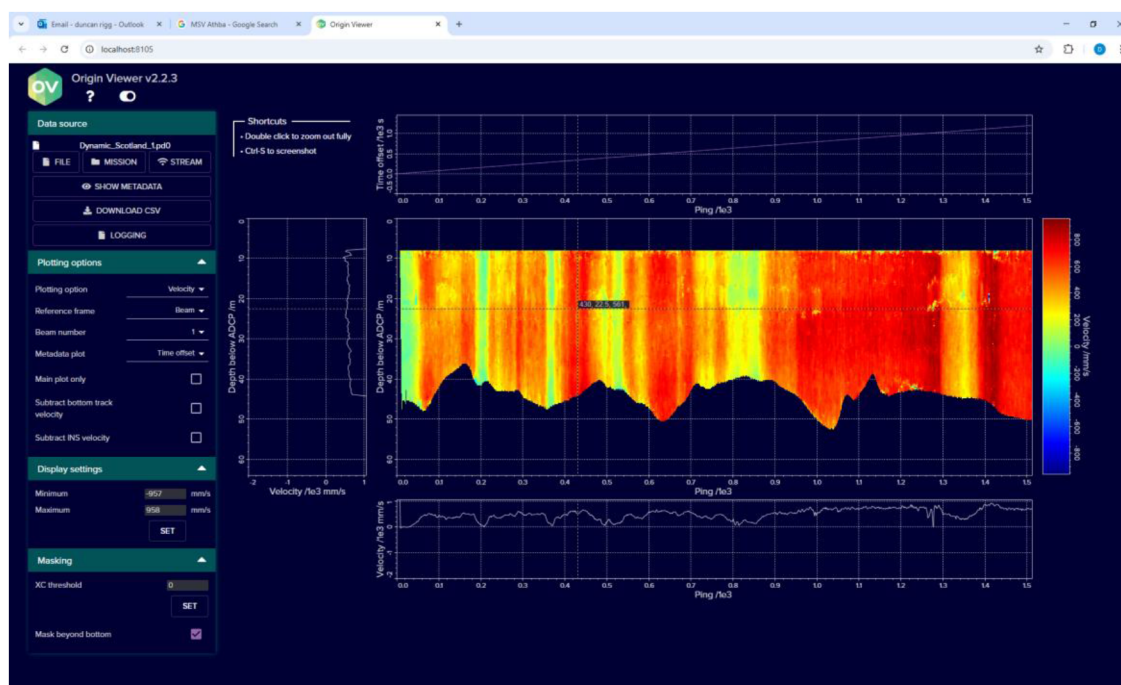


Figure 2—Acoustic Doppler Current Profiling (ADCP) data gathered using Sonardyne's SPRINT-Nav DP. Waterfall plot represents water current velocity in mm/s ranging between 800 to -800 mm/s in system unit reference frame

What are the limitations?

SPRINT-Nav DP uses a 375kHz Sonardyne Syrix DVL. This frequency of DVL has been selected as it offers an optimal balance of signal resolution and range. Although the system has been proven to work in water depths of 225m, seabed composition can impact the maximum operational altitude. A hard rocky seabed offers the most reflective surface and therefore provides the maximum working depth. Soft, muddy seabeds absorb the acoustic pulse and therefore a reduced operating range will be achieved. The results of the maximum altitude testing are shown in Figure 3. The signal to noise ratio (SNR) and cross-correlation, an indicator of signal quality, can be seen to both degrade as the vessel passes over 225m altitude. The result of this test was in line with Sonardyne's expectations for the system. As the seabed was sandy it is expected that the system would potentially be able to work in deeper water where there was a hard seabed.

As with other acoustic systems, SPRINT-Nav DP is susceptible to interference from other noise sources, however unlike many other acoustic positioning systems the frequency range is outside common ship borne noise (0-16 kHz)². Aerated water will inhibit the signals as they propagate through the water column, so therefore, much like HPR systems, careful installation location is required. In addition, thruster management may be required to ensure thruster-wash (aerated water) is not directed over the sensor.



Figure 3—DVL signal drop-out at 225m altitude at system time 7289. The signal drop out is confirmed by an decrease in the SNR, and velocity and range drop-out. Note range is slant range; equates to 225m altitude with 30 degree tilt on DVL transducers

Additional Features

SPRINT-Nav DP offers various operational benefits over a stand-alone PRS. In addition to providing an independent vessel position, the system can also be used to provide high-grade, pitch roll and heading data that can be used for survey operations or as a third gyro into the DP.

INS systems tend to require a settling period after initialization. SPRINT-Nav DP includes a battery that will maintain power to the system for 5-minutes. This assists in preventing a settling time occurring after periods of power outages, allowing the system to be operational quicker.

Trials Data

SPRINT-Nav DP has been used on various offshore trials during the development of the system with the first offshore use as an input to DP in 2019. One trial to note is that onboard the MV Brouwersgracht, a heavy lift vessel operated by Spliethoff. The vessel conducted maximum altitude testing when operating offshore Singapore. On completion of maximum altitude testing, an FMEA was conducted with SPRINT-Nav DP connected to the GE Seastream DP system. The position plot during the 20m FMEA box maneuver is shown in Figure 4.

During extended DP operations conducted at altitudes of 225 meters or less, the DVL-INS system demonstrated strong performance, maintaining sub-2-metre positional error (difference from the truth reference) over a continuous 24-hour period. This performance illustrates the robustness of DVL bottom-lock aiding in constraining INS drift, even when dynamic motion is limited.

In higher-dynamic scenarios, such as during transits, the system's performance improved further. The DVL-INS solution achieved a drift of just 0.12% of distance travelled (CEP50), relative to the ground-truth Differential GNSS (DGNSS) position. This low drift rate highlights the effectiveness of DVL-INS fusion during underway conditions, where frequent heading changes and sustained motion provide continuous opportunities for INS error correction. The result is a highly stable and accurate surface navigator, even in the absence of real-time GNSS updates.

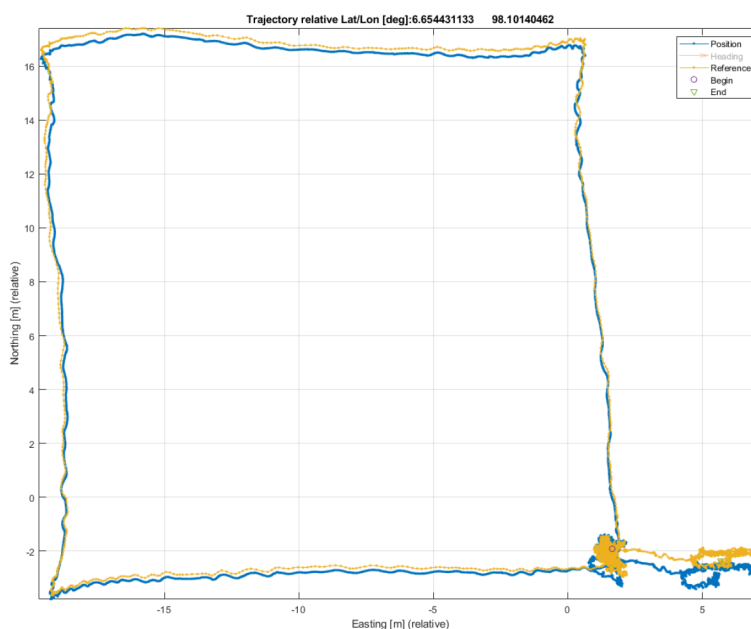


Figure 4—SPRINT-Nav DP FMEA 20m box maneuver conducted in 225m water depth results onboard MV Brouwersgracht. DGNSS in yellow, SPRINT-Nav DP in blue

Conclusion

Sonardyne's SPRINT-Nav DP represents a significant advancement in vessel dynamic positioning (DP) capability, offering a reliable and versatile alternative to conventional DP reference sensors. Leveraging integrated inertial and Doppler velocity technologies, the system enables accurate subsea positioning without the need for additional deployed infrastructure, eliminating vulnerabilities such as sensor masking from surrounding structures.

Extensive field validation has demonstrated that SPRINT-Nav DP delivers robust and consistent positioning performance in water depths of up to 225 meters, supporting extended operations across a wide range of continental shelf environments. This capability provides operators with greater operational flexibility, particularly in complex offshore scenarios where minimizing in-water equipment is advantageous.

When deployed alongside other PRSs, SPRINT-Nav DP enhances system redundancy and situational awareness. It supports the detection of GNSS spoofing and contributes to real-time environmental awareness by providing water current data. As offshore operations become more complex and data-driven, SPRINT-Nav DP offers a future-ready solution for maintaining reliable vessel positioning and operational continuity.

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